

Hamilton Award Proposal

Alexander Marshall

February 27, 2026

Table of Contents

1. Other Sources of Funding	1
2. Summer Funding Plan	2
3. Letter of Recommendation	3
4. Curriculum Vitæ	4

Other Sources of Funding

I am a Royster Doctoral Fellow. Presently, I am scheduled to spend the Summer of 2026 as a non-service semester. If I do not receive the Hamilton award, I will receive \$5000 as a stipend from the Graduate School. The Royster Fellowship is flexible, and if I do receive the Hamilton Award, I will elect to repurpose this funding to advance my research at a later date so I can receive the full benefits of the Award.

Summer Funding Plan

I will use the Hamilton Award to enable full-time commitment to research this summer, allowing me to advance the goals of my project more quickly.

Tissues and individual cells are poroelastic, consisting of a background matrix suffused by a solvent carrying solutes of diverse sizes. In vivo, tissues undergo dynamic stresses, causing deformations which induce advective flows. Size-dependent transport of proteins and vesicles ensues; solutes larger than the pore size of the tissue are greatly hindered by interactions with the mesh, while smaller solutes move more freely. Experiments examining the macroscale dynamics of poroelastic media demonstrate enhanced concentration of large solutes in cyclically strained hydrogels, but there remains a dearth of studies probing the intricate interactions between solutes, pore size, stress gradients, and microfluidics during dynamic strains in living tissue and cells, and thus the roots of these emergent behaviors remain unclear.

My work to date has consisted of tracking single nanoparticles in strained poroelastic media, providing the first live insight into the nanoscopic dynamics of these systems. I have shown that both static and dynamic strains increase long-term localization and decrease transport compared to unstrained controls, while dynamic strain induces asymmetric advective transport of large solutes during compression and relaxation, creating a pumping effect. My work to date has been completed with the goal of applying identical methodologies to live-cell imaging in order to understand how solute distribution in the nucleus, a heterogeneous poroelastic environment often subjected to dynamic strains, drives gene transcription.

This summer, I will extend the study to living systems by tracking transcription factors (TFs) in live cells to understand how strains impact their transport and distribution, ultimately altering gene expression. By connecting imposed dynamic strains to the redistribution of TF I will uncover a novel mechanism for how cells can respond to external stress. Doing so will require single particle tracking of fluorescently labelled TFs in live cellular nuclei subject to controlled strains.

To accomplish this, an existing cell line will be treated with non-toxic HALO-tag solution to fluorescently label transcription factors. The cells will be imaged on a light sheet fluorescence microscopy system of my own design operating with low phototoxicity within a climate-controlled incubation system, allowing for extended experiments on healthy cells. Precise strain will be applied during live imaging using a $20\mu\text{m}$ spherical indenter mounted on a piezoelectric-actuated atomic force microscope cantilever. The strain system is mounted on a micro-manipulation stage, allowing for precise positioning. Fluorescently-labelled TFs will be imaged using highly inclined laminated optical sheet fluorescence microscopy (HILO) to acquire high-framerate time series of the particles. Individual TFs will be tracked using TrackMate, an open-source state-of-the-art tracking platform, and the resulting tracks will be analyzed via msdalyzer (an open-source MATLAB plugin for single particle tracking) and personal MATLAB scripts. As an extension, a unique microscope enabling simultaneous atomic force microscopy and HILO will be used to produce simultaneous particle tracking and microrheological data.

I have applied this approach to successfully track comparable nanoparticles in strained poroelastic hydrogels at up to 800 frames per second; furthermore, recent collaborations with Dr. Kerry Bloom of UNC Biology have demonstrated single particle tracking of HALO-tagged proteins in the nucleus. Investing three years of my work has led to the development of a system with capabilities necessary for this undertaking; along with my recent experiments in hydrogels, this positions me to uncover new phenomena in living cells this summer.



February 27, 2026

To Whom It May Concern,

I am delighted to have the opportunity to enthusiastically recommend Alex Marshall for the Hamilton Award. I have come to know Alex over the past four years since he joined our group as a graduate student in 2022, even before enrolling at UNC. Alex started in the group with a bang – immediately taking on one of our most challenging experiments – developing a new volumetric light sheet microscopy system for our biophysical and biochemical studies of live cells. We have emerged from an unusual period in the lab, with our two senior graduate students working off site (and out of state) this past year while we welcomed into the lab three new graduate students, including Alex. This has meant that Alex and his cohort have not had the mentorship of senior students during their initial ramp up period in the lab. This cold launching has been embraced enthusiastically by Alex, and he has stepped up mightily to take on his challenging project.

Within the research conducted in my own group, Alex has chosen to step into our most challenging project, advanced instrumentation for studying the biophysics of live biological cells. Living cells and tissues are complicated materials that challenge our understanding of even the most basic processes in man-made materials. They are like sponges – they have a matrix of molecules, proteins and polymers, all submerged in a fluid that pervades the sponge. This fluid carries smaller proteins and particles that can move through the sponge – depending on their size. If they are smaller than the holes in the sponge – the particles move freely and flow with the movement of the fluid. If they are larger than the sponge holes, the particles get stuck. In between sizes move more or less freely with the larger particles being slowed down by the sponge. In materials science, these materials are called poroelastic. They are elastic because the material can deform as it is squeezed. They are called “poro – “ because of the pores in the material that allow fluid to flow. The interesting thing about these materials is that, of necessity, if an external force is applied - the material will deform and the fluid will flow. In living tissues and cells, the “sponge” is made up of biopolymers such as actin filaments, microtubules, extracellular matrix, and within the cell nucleus, DNA. The fluid that pervades the biomatrix carries small molecules, proteins of various sizes and vesicles. These move within the biomatrix according to their size, with the largest of them having their movement severely restricted. Within the nucleus, it is the movement and distribution of proteins and their complexes called transcription factors that dictate the generation of specific proteins through transcription. Our hypothesis is that dynamic strains on tissues and cells results in the deformation of the nucleus which causes fluid flow that redistributes transcription factors (TF) and therefore alters the proteins that are generated in the cell. This would be a novel insight into how cells respond to external mechanical stresses such as occur in muscles, in joints, and traumatic injury.

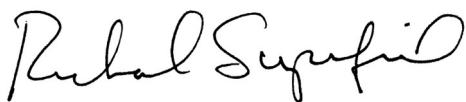
Remarkably, there are very few studies of these phenomena in man-made materials, and even fewer in live cells. This is the heart of Alex’s studies. He has spent the first three years of his lab work developing a novel light sheet microscope for his studies, building the data analysis tools to probe the data he acquires and developing the hydrogel specimens to study stress-induced fluid flow and solute redistribution. These are the first studies at the single particle level of this phenomenon, unique to poroelastic materials. Alex has uncovered the first signs of asymmetric transport during dynamic compressions of hydrogels which will provide the first molecular scale understanding of strain pumping of proteins. This will have a dramatic impact on tissue engineering, on the treatment of joint disorders and provide a new mechanism for the cell-level sensing of mechanical stresses.

Coming from a physics mindset, Alex seeks to understand this process through a predictive theoretical physics model that combines an understanding of physical chemistry and biology. To create this model, he first needs to acquire rapid volume scans of hydrogel materials during dynamic strains, and executing time-synchronized single particle tracking of thousands of mobile nanoparticles. This he has accomplished with little help from senior staff. He is now putting together his work into two publications on novel poroelastic phenomena in hydrogel materials. These will be high impact manuscripts. Alex has planned his studies to be immediately transferrable to experiments on live cells, and is actively working with our collaborator Prof. Kerry Bloom to begin those experiments. He has also started analyzing data from the Bloom lab of the mobility of individually labeled histone proteins to understand their dynamics within the nucleus. His next steps will be to apply dynamic strain to living cell nuclei while tracking the histones, labeled transcription factors and other proteins.

Alex represents a new generation of scientist that knows no disciplinary boundaries, but brings instincts for framing understanding from his starting mindset: physics. Merging fields from physics to materials science to physical chemistry to biology and instrumentation, Alex is endeavoring to do it all! He has accomplished the first leg of this journey: to develop advanced volumetric microscopy methods and instrumentation, and to execute these on hydrogel materials. Alex has accomplished all of this single handedly in my group. While in past years my starting students have come into the lab with the mentorship of senior students, Alex has not had this luxury. I am thrilled that he is bringing his eclectic mind to my group.

Alex is bright, creative and an incisive thinker. Most important, he is highly coachable and is taking personal responsibility for his training in optics, microscopy and cell biology. There is no more important trait for a researcher than their devotion to learning and to teaching themselves. The most successful students are voracious in reading the scientific literature and identifying their own knowledge gaps – and filling these in rapidly. Alex is wonderful at doing these. On the personal side, he is a collegial lab mate with a ready smile and laugh, and generous with his time for mentoring and collaborating. He has a long record in mentoring and teaching undergraduate students that is most impressive. He has assembled a mini-group around him, coaching 9 undergraduates in total, with 4 of them currently active in the group. I am especially impressed with his appreciation of the ways in which he has been mentored across generations of family and friends, and how that has instilled in him a deep enthusiasm - and a sense of obligation – for his outreach activities. He is an enthusiastic presenter at our many presentations to high school and undergraduate students, and to the booth we set up at the annual NC Science Festival.

Alex is an outstanding example of what the Hamilton Award should support: a clear, physics-based thinker tackling challenges at the forefront of biology while also deeply concerned with the world around him. I give him my highest recommendation.



Richard Superfine
Department of Applied Physical Sciences
Taylor-Williams Distinguished Professor

Alexander Marshall | Physicist

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Royster Fellow | Doctoral Candidate | UNC-Chapel Hill

Education

Doctorate of Philosophy in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2027

Royster Doctoral Fellow | Exams Passed | All But Dissertation

Master of Science in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2025

Completed *en route* to PhD

Physics B.S. | Philosophy B.A. | Mathematics Minor

Winston-Salem, NC

Wake Forest University

2022

GPA 3.91 | *Summa Cum Laude* | Honors in Physics | Stamps Scholar

Academic Projects

Doctoral Thesis.....

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Masters Thesis.....

Physiological Strains in Poroelastic Media Induce Asymmetric Transport

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Undergraduate Honors Thesis.....

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator. Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Additional Research.....

Chromatin Structural Evolution Alters Histone Transport Dynamics **Chapel Hill, NC**

UNC-Chapel Hill, Dr. Kerry Bloom

2025

Tracking and analyzing the trajectories of HALO-tagged histones during metaphase and interphase to probe structural changes occurring in chromatin.

Enabling Low-Resource Microscopy

Chapel Hill, NC

UNC-Chapel Hill, Dr. Richard Superfine

2023-Present

Supervised a primarily undergraduate-driven effort to develop cutting edge microscopy techniques for use in low-research environments. Undergraduates won prizes at annual UNC MakerFest competition for two consecutive years.

Employing Advanced LSFM for Enhanced Imaging

Chapel Hill, NC

UNC-Chapel Hill, Dr. Richard Superfine

2022-Present

Engineered an advanced optical system to conduct new methods of LSFM, such as HILO and HIST. Due to software rot, had to rebuild system from the ground up, involving extensive design and engineering. Also implemented new optical modules and DAQ-board mediated software to enable sub-millisecond automated control of optical hardware.

Synthesizing Chalcogenide Crystals for Quantum Computing

Winston-Salem, NC

Wake Forest Nanotech Center, Dr. David Carroll

2021

Assisted with the synthesis of chalcogenide crystals being investigated as a potential substrate for quantum computing.

Neurochemical Addiction Mechanisms In the Nucleus Accumbens

Winston-Salem, NC

Wake Forest School of Medicine, Dr. Evgeny Budygin

2019-2021

Conducted cyclic voltammetry on rats *in vivo* and mouse brain slices *in vitro* to explore the role of neurotransmitters in addictive behaviors.

Cultural Impact of Oenology in the Peloponnese

Winston-Salem, NC

Wake Forest Department of Anthropology, Dr. Karen Friederic

2019

Conducted a month-long anthropological study in Nemea, Greece on how the economic significance of the viticultural industry impacts local culture. I found that extreme seasonal workloads intersect with traditional gendered domestic labor expectations to inhibit female career growth in this industry.

Cancer and Postoperative Pain

Winston-Salem, NC

Wake Forest Baptist Medical Center, Dr. Christopher Peters

2018

Prepared solutions, assisted with operations on research subjects, and conducted fluorescence microscopy on brain slices from rats to ascertain the distribution of optogenetic proteins in neurons after treatment with a viral vector.

Scholarships and Grants

Royster Doctoral Fellowship

Chapel Hill, NC

Royster Foundation, UNC-Chapel Hill Graduate School

2022-2027

Five-year doctoral recruitment fellowship. Provides extradepartmental funding for the majority of a five year doctoral program.

Stamps Scholarship

Winston-Salem, NC

Stamps Foundation

2018-2022

Four-year scholarship awarded for academic merit. Covered all costs of attendance to Wake Forest University, including room and board. Also provided a technology grant and semesterly stipend.

Summer Research Project Grant

Winston-Salem, NC

Wake Forest University

2019

Granted for summer anthropology project studying the cultural impact of viticulture in the Peleponnese.

Publications

Strain alters large solute transport in poroelastic media

Chapel Hill, NC

Manuscript In Preparation, UNC-Chapel Hill

Anticip. 2026

Alexander Marshall, Richard Superfine

Dynamic strains drive asymmetric advection in poroelastic media

Chapel Hill, NC

Manuscript In Preparation, UNC-Chapel Hill

Anticip. 2026

Alexander Marshall, Richard Superfine

Technical skills

Laboratory Safety Manager

Chapel Hill, NC

UNC-Chapel Hill, Dept. of Environmental Health and Safety

2022-Present

Responsibilities include writing the lab safety plan, ensuring compliance during day-to-day operations, ensuring safety training compliance, disposing of hazardous waste, coordinating with EHS to maintain a safe workplace, and conducting regular inspections of critical safety equipment.

Programming: MATLAB, LabVIEW, Python, L^AT_EX, Git, Arduino

Software: Microsoft Office, Spot-On, μ Manager, ImageJ, FIJI, autoCAD

Data Analysis: Particle Tracking, Image Analysis, Stochastic Processes, Machine Vision, Image Deconvolution

Microscopy: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Confocal Microscopy

Optical Systems: Custom Optical Design, Laser Systems, Spatial Light Modulation, Remote Focusing, Accusto-Optic Tunable Filters, 4-F System Design

Electronics: Multimeter proficiency, soldering, custom electronics design, multimodal control design, system timing control, DAQ-board interfacing, system control software design, Arduino-based control, Raspberry-Pi based control

General Lab Proficiencies: Chemical protocols, highly toxic chemical safety, fume hoods, cell culture, power tool use, general workshop skills

Machine Shop: Took a semester-long machine shop course at Wake Forest University in 2021.

Certified as departmental machine shop operator following this course.

Talks and Symposia

Physiological Strains in Poroelastic Media Induce Asymmetric Transport <i>Master's Defense, UNC-Chapel Hill</i>	Chapel Hill, NC November 17, 2025
Strain Alters Anomalous Diffusion in Poroelastic Media. <i>Symposium Talk, Elon University</i>	Elon, NC October 1, 2025
Diffusion in Strained Poroelastic Media <i>Thesis Proposal, UNC-Chapel Hill</i>	Chapel Hill, NC September 16, 2025
Generating a 3-D Lattice of Photonic Spheres <i>Honors Thesis Defense, Wake Forest University</i>	Winston-Salem, NC May 2022
Generating a 3-D Lattice of Photonic Spheres <i>Symposium Talk, Wake Forest University</i>	Winston-Salem, NC May 2022

Conferences

Strain-Driven Transport In Poroelastic Media <i>Biophysical Society Meeting, Biophysical Society</i>	San Francisco, CA February 21, 2026
Strain-Driven Transport In Poroelastic Media <i>13th Carolina Biophysics Symposium, UNC-Chapel Hill</i>	Chapel Hill, NC November 6, 2025
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, Duke University</i>	Durham, NC September 15, 2025
Strain-Driven Transport In Poroelastic Media <i>Royster Global Conference, Royster Society of Fellows</i>	Chapel Hill, NC October 21, 2025
Strain-Driven Transport In Poroelastic Media <i>Triangle Soft Matter Workshop, Duke University Dept. of Physics</i>	Durham, NC May 12, 2025
Strain-Driven Transport In Poroelastic Media <i>8th International Soft Matter Conference, Soft Matter Association of the Americas</i>	Raleigh, NC August 2024
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting</i>	Raleigh, NC September 23, 2024
Probing Structural and Force Dynamics during Phagocytosis <i>Triangle Cytoskeleton Meeting</i>	Raleigh, NC September 25, 2023
Various topics <i>Annual Colloquium, UNC Applied Physical Sciences Department</i>	Chapel Hill, NC Annually since 2022
Stemps Scholars Convention <i>Attendee, Georgia Tech</i>	Atlanta, Georgia Fall 2019
The Cultural Impact of Oenology in the Peleponnese <i>Undergraduate Research Day, Wake Forest University URECA Center</i>	Winston-Salem, NC September 2019

Teaching

Research Mentor

Superfine Lab, UNC-Chapel Hill

Chapel Hill, NC

Fall 2022-Present

Mentored a diverse group of undergraduates, helping them to develop as researchers while working on various projects. Persistent efforts at outreach drove undergraduate recruitment and engagements in the lab, leading to a dramatic expansion of the lab's engagement with undergraduates. The research activities of select students I advised are summarized in the mentorship section.

Teaching Assistant

Physics I, Wake Forest University

Winston-Salem, NC

Fall 2021

Due to COVID-19 restrictions, the instructor of record was unable to acquire a work visa and could not enter the United States, requiring the assumption of additional responsibilities.

Teaching Assistant

Physics I, UNC-Chapel Hill

Chapel Hill, NC

2023-2024

Primarily acted as a lab instructor across three semesters. Responsible for day-to-day administration of two 110 minute 75 student lab sections which met biweekly.

Teaching Assistant

Various Physics Courses, UNC-Chapel Hill

Chapel Hill, NC

2023-2024

Served as a TA for various other courses. Topics covered included Thermodynamics, Quantum Physics II, a first-year seminar centered on the philosophy of science, working in the departmental assistance center, and evaluating student performance.

Departmental Tutor

Physics I and Physics II, UNC-Chapel Hill Naval Armory

Chapel Hill, NC

Fall 2023-Spring 2024

Provided physics tutoring for ROTC-enrolled students as part of a departmental initiative.

Academic Tutor

STEM Topics, Private Clients

Various Locations

2018-Present

Worked one-on-one with students of high-school and college age on topics ranging from high school chemistry to Physics Olympiad preparation.

Mentorship

Ashleigh Smith

Undergraduate Mentee, Anomalous Diffusion

Chapel Hill, NC

2024-Present

Ashleigh created a PAAm gel protocol to meet complex experimental requirements for a poroelastic substrate in an isolated chamber. As she has matured as a researcher, this work has segued into a more autonomous project studying the wicking effect of poroelastic media.

Mason Miller

Undergraduate Mentee, Anomalous Diffusion

Chapel Hill, NC

2025-Present

Mason designed a 3D printed strain control system and ran many strain experiments in poroelastic media. Mason is now working to characterize the rheologic properties of poroelastic media during strains via atomic force microscopy.

Bradley Mankus

Undergraduate Mentee, Anomalous Diffusion

Chapel Hill, NC

2025-Present

Bradley has improved upon the PAAm gel preparation protocol, has also begun work on developing novel microfluidics channels for testing the effect of advective flows on diffusion in poroelastic media.

Shivam Patel*Undergraduate Mentee, Anomalous Diffusion*

Shivam has worked to expand my data analysis codebase, and is currently working to simulate strain-induced advection in poroelastic media.

Chapel Hill, NC*2025-Present***Kautilya Chopra***Undergraduate Mentee, Anomalous Diffusion*

Kautilya created python-based demos of brownian motion from first principles. He then expanded the simulation to model particle trapping and other dynamics typical of diffusion in poroelastic media.

Chapel Hill, NC*Summer 2025***Akshay Elango***Undergraduate Mentee, Low-Resource Microscopy*

Akshay built a 3-D printed transmission microscope based on the OpenFlexure system. His reserch is currenty on hiatus while he works in an industrial internship, but he will return upon its completion.

Chapel Hill, NC*2023-Present***Anya Frazer***Undergraduate Mentee, Low-Resource Microscopy*

Anya expanded upon the OpenFlexure Microscope system, enhancing the capabilities and software of the fluorescence scope, including upgrading it to conduct single particle tracking of nanoscopic particles. She later developed a 3-D printed molecular stamper designed to manufacture Janus particles at dramtically reduced cost in low-resource settings.

Chapel Hill, NC*2024-2025***Eric Chen***Undergraduate Mentee, Low-Resource Microscopy*

Eric built a 3-D printed fluorecence microscope based on the OpenFlexure system. He also designed and manufactured his own filtersets for this system from bulk sheets of Schottglass to minimize costs to advance the low-resource intensity goals of the project.

Chapel Hill, NC*2023-2024***Joshua Adekunle***Undergraduate Mentee, Low-Resource Microscopy*

Josh built a 3-D printed low-cost transmission microscope based on OpenFlexure.

Chapel Hill, NC*2023-2024*

Outreach

Open Flexure Microscope and Molecular Stamper*UNC-Chapel Hill*

Initiatives planned for building microscopes in low-resource highschoools in Eastern NC.

Chapel Hill, NC*2023-Present***North Carolina Science Festival***Morehead Planetarium and Science Center*

Each year, during this public outreach event the Superfine research group conducts science demos and research talks targeted at general audiences.

Chapel Hill, NC*2023-Present***Hospital Volunteer***Wake Forest Baptist Medical Center*

Volunteered in various patient-facing positions for over 300 hours.

Winston-Salem, NC*2015-2017*

References

Richard Superfine, PhD*superfine@unc.edu, (919) 432-4381*

Taylor-Williams Distinguished Professor, UNC-Chapel Hill

Doctoral Advisor*2022-Present*

Kerry Bloom, PhD
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Thad L. Beyle Distinguished Professor, UNC-Chapel Hill

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Vice Provost, Wake Forest University

Martin Guthold, PhD
guthold@wfu.edu, (336) 758-4977
Physics Dept. Chair, Wake Forest University
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Research Collaborator
2025-Present

Doctoral Committee Member
2025-Present

Undergraduate Advisor
2019-2022

Personal Reference
2010-Present

Languages

English: Fluent	<i>Native Speaker</i>
Contemporary Greek: Intermediate	<i>Conversational speaker.</i>
Classical Latin: Intermediate	<i>Seven years of formal instruction</i>